Boolean degree one functions on the Grassmann scheme — easy

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Filmus and Ihringer (Boolean degree 1 functions on some classical association schemes) classified all Boolean degree 1 functions on the Grassmann scheme $J_2(n,k)$ whenever $\min(k,n-k) \geq 2$; here $J_2(n,k)$ is the collection of all k-dimensional subspaces of $\mathbb{F}_2^n$. If $f\colon J_2(n,k) \to \mathbb{R}$ is Boolean ($f(L) \in \{0,1\}$ for all L) and has degree 1 (can be written as a linear combination of functions of the form $p_x(L) = [x \in L]$) then either f or $1-f$ has one of the following forms:

$$0, [x \in L], [y \perp L], [x \in L \text{ or } y \perp L],$$

where $x, y \neq 0$, and in the last case, $x \not\perp y$.

Problem 1. *Determine all degree* 1 *functions* $f\colon J_2(n,k) \to \mathbb{R}$ *which are* $\{0,1,2\}$*-valued, where* $\min(k,n-k) \geq 2$.

(If it helps, you can assume that $\min(k,n-k) \geq t$ *for some small constant* t*.)*

If instead of $\mathbb{F}_2$ we work with a larger finite field $\mathbb{F}_q$, then there are strange examples that arise in $J_q(4,2)$. For small q, they disappear already in $J_q(5,2)$, and the theorem above extends whenever $\max(k,n-k) \geq 3$ and $\min(k,n-k) \geq 2$. Ihringer shows that this happens for every q, with "3" replaced by a (huge) constant depending on q. This explains how t might pop up.

When working over the Boolean cube, the following functions arise:

$$0, x_i, x_i + x_j, 1, 1 + x_i, 1 - x_j, 1 + x_i - x_j, 2, 2 - x_i, 2 - x_i - x_j.$$

We suspect that the case of the Grassmann scheme is similar, where we also make use of the fact that if $x \not\perp y$ then $[x \in L]$ and $[y \perp L]$ cannot hold at the same time. (This can be used to guess the answer.)

The recent survey "Boolean degree one functions on the Grassmann scheme" of Filmus gives a proof strategy that should succeed here as well, with a little bit of work. (The hard part of that paper is a proof of Ihringer's result, but this won't be needed here.)